

MATH9102

DIMENSION REDUCTION

Reference: Field, Miles and Field Discovering Statistics Discovering
Statistics with R

Dimension Reduction

Process of reducing the number of variables (or features) in a dataset while retaining as much of the important information as possible.

This is often done to simplify analysis, improve computational efficiency, or prevent overfitting in predictive models.

Why are we looking at dimension reduction?

The statistical researcher strives to avoid two major obstacles:

- Using a single score/measure from a multi-item measure in which there is great heterogeneity;

OR

- Using several scores/measures that are highly correlated or unreliable.

The goal is to obtain the “right” number of scores (and it might be one).

Collinearity

Occurs when two or more independent variables contain strongly redundant information

- If variables are collinear, they are essentially measuring the same thing
 - There is not enough distinct information in these variables for inferential statistics to operate

Why is this a problem?

- You are introducing bias
 - Over-representing a concept
- Increasing possibility of a Type I error

If we conduct inferential statistics with collinear variables, then the model will produce unreliable results

Collinearity

Need to check for collinearity by examining a correlation matrix that compares your independent variables with each other.

A correlation coefficient above 0.8 suggests collinearity ***might*** be present

Collinearity – What can we do about it?

1. Delete all but one of the collinear variables from the model
 - You could choose which variables you feel are most representative of the concept you are interested in
 - It will act as a surrogate or proxy for the concept
 - E.g. number of previous criminal offences as a predictor of reoffending
 - May weaken the fit of your model
2. Combine them into an index mathematically (e.g. multiplying, adding etc.)
 - This must make sense theoretically
 - E.g. Select a range of items
 - The Retail Price Index is calculated by taking a sample of goods and services a typical household buys and items weighted according to the amount spent on them

Collinearity – What can we do about it?

3. Estimate a **latent** variable using an appropriate data reduction technique
 - Latent variable: a variable that is not directly observed but is inferred from other variables that are observed (measured)
 - How do we estimate?
 - Factor analysis (FA)/Principal Component Analysis (PCA)
 - Construct indices by methodologically assessing contributions of a range of variables so that a combined measure can be extracted and used.

Why dimension reduction?

We know

- We are using variables as proxy measures for real world concepts
- Multicollinearity can cause problems when building predictive models

We aim

- For parsimony in our models (fewest number of predictor variables)
- To avoid two major obstacles:
 - Using a single variable created from a multi-item measure in which there is great heterogeneity;
or
 - Using several variables that are highly correlated or unreliable.

Our Goal is To

- Obtain the “right” number of variables to represent a concept
 - (and it might be one).

Everything that you need to know about Dimension Reduction

The basic issue is the **degree of correlation among a set of items**.

We want to test for **clusters** of variables or measures.

- That is:
 - To see whether different measures are tapping aspects of a common dimension.
 - For example, people may respond similarly to questions about income, education, and occupation, which are all associated with the latent variable socioeconomic status.

We expect to find “*clumps*” of items sometimes

- These are called **factors** (if doing factor analysis) **or components** (if doing principal components analysis).

Everything that you need to know about Dimension Reduction

In every dimension reduction:

- The number of factors/components will be the same as the number of variables used in the dimension reduction.
- Each factor/component captures a certain amount of the overall variance in the observed variables.
- The factors/components are always listed in order of how much variation they explain.
- We make decisions about which factors/components are useful to us
 - By examining the amount of variance in the observed variables each factor/component can be considered to explain.

What are we doing?

We are theorizing that there is a single individual *latent* concept, or a small number of concepts, which *manifests* through several different aspects

- These manifestations are collected in variables in the dataset

We are trying to distil the underlying concept(s) from its manifestations

- But how many dimensions does a concept have?
- How should they be represented?

Why dimension reduction?

Many phenomena cannot be measured directly

- No measurement scale exists
 - E.g. social deprivation, cost of living, consumer confidence etc.
- These are **latent** concepts
 - They are inferred through observable indicators which can be measured
- Constructs like "intelligence" or "satisfaction" are latent because they cannot be measured directly.
 - Instead, they are inferred through observable (manifest) indicators, like test scores or survey responses.

Why dimension reduction?

To derive/expose these latent concepts

- We use ***manifest variables*** which store the observable indicators which measure aspects of the ***latent variable***

We construct a measurement scale from the manifest variables

- E.g.
 - To construct a measure of intelligence we could use the following manifest variables:
 - Result on an IQ test as an indicator of ability to perform particular types of problem solving and related cognitive skills
 - Scores from tests measuring verbal reasoning, mathematical ability, spatial reasoning, or logical reasoning.
 - Academic grades in subjects like mathematics, science, or language.
 - Scores on creativity assessments like Torrance Tests of Creative Thinking (TTCT).
 - Scores on tasks or questionnaires evaluating emotional intelligence or social reasoning.

When do we use it?

We may be:

- Interested in creating an agreed measure that will allow us to track changes over time
- Interested in exploring particular dimensions of a concept
- Pragmatically, be interested in reducing the number of variables we need to work with in an analysis

All the above are not mutually exclusive

Approaches?

Factor Analysis

- Exploratory
 - Used in early stages of research to gather information about interrelationships between variables
 - e.g. you have created an instrument with multiple questions, and you want to test how well they measure concepts of interest
- Confirmatory
 - Used later in research to confirm hypotheses about the structure underlying a set of variables

Principal Components Analysis

- Used when working with a set of uncorrelated variables which it has been well established that they measure the same underlying component (s).

What are we looking at?

Goal of the Process

- Start with a large set of variables.
- Reduce them to a smaller set of new variables.
- These new variables should still capture most of the information in the original dataset.

How We Measure “Information”

- **Variability within variables**
 - *Variance*: how much each variable's values differ from its mean.
- **Co-variation across variables**
 - *Covariance / Correlation*: how strongly two variables change together.

Key Idea

- If variables share patterns of variability or move together (high covariance), the information they contain overlaps.
- Techniques like PCA summarise these shared patterns into fewer, more informative components.

Factors/Components?

Factors / Components = Latent Variables

- In this context, a **factor** or **component** is the same as a **latent variable** (*also called a **construct***).

Terminology (All Mean the Same Thing)

- **Construct = Latent Variable = Factor / Component**

Getting started

Need to be working with interval/ratio or ordinal data

- Produce basic descriptive statistics look for missing or outlier values and take appropriate action.

All the variables should measure the construct in the same direction.

- The direction does not matter the important thing is that all the questions score in the same direction.
- If they differ, then recode to ensure that the scales are the same.
 - E.g.
 - *"I find it easy to stay motivated when working on long-term projects."* (Positive)
Responses:
1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
(Higher scores indicate higher motivation.)
 - *"I often feel overwhelmed and give up easily when tasks seem too complex."* (Negative)
Responses:
1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
(Higher scores indicate higher difficulty managing complexity, which reflects lower motivation.)

The negative-direction question needs to be **reverse-coded** so that higher scores consistently indicate higher motivation.

Recoding Example

"I find it easy to stay motivated when working on long-term projects." (Positive)

Responses:

1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
(Higher scores indicate higher motivation.)

"I often feel overwhelmed and give up easily when tasks seem too complex." (Negative)

Responses:

1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
(Higher scores indicate higher difficulty managing complexity, which reflects lower motivation.)

```
# Create dataframe
responses <- data.frame(
  motivation_positive = c(4, 5, 3, 2,
1),
  overwhelmed_negative = c(5, 4, 3, 2,
1))

# Reverse-coding the negative-
direction question
responses$overwhelmed_reversed <- 6
- responses$overwhelmed_negative

# Check the recoded data
print(responses)
```

Example (From Andy Field)

Factor analysis could be used to identify the “core” characteristics (out of a potentially large number of characteristics) that make a person popular.

Candidate characteristics:

- Time spent talking about the other person (Talk 1 – a relatively positive trait)
- Level of social skills (Social Skills)
- How interesting a person is to others (Interest)
- Time spent talking about themselves (Talk 2 – a relatively negative trait)
- Selfishness (Selfish)
- The person’s propensity to lie about themselves (Liar).

Example

Looking at how the concepts relate to each other.

	Talk 1	Social Skills	Interest	Talk 2	Selfish	Liar
Talk 1	1.000					
Social Skills	.772	1.000				
Interest	.646	.879	1.000			
Talk 2	.074	-.120	.054	1.000		
Selfish	-.131	.031	-.101	.441	1.000	
Liar	.068	.012	.110	.361	.277	1.000

Factor 1

Factor 2

Look for correlations in the Correlation matrix.

In Factor Analysis and PCA we look to reduce the R-matrix into smaller set of uncorrelated dimensions.

Looking for correlations above 0.3.

We seem to have 2 clusters.

Example

Looking at what concepts relate to sociability and consideration traits in people

	Talk 1	Social Skills	Interest	Talk 2	Selfish	Liar
Talk 1	1.000					
Social Skills	.772	1.000				
Interest	.646	.879	1.000			
Talk 2	.074	-.120	.054	1.000		
Selfish	-.131	.031	-.101	.441	1.000	
Liar	.068	.012	.110	.361	.277	1.000

Factor 1

Factor 2

Social Skills is strongly correlated with Time spent talking about the other person

How interesting a person is to others is highly correlated with both of these.

These are positive traits.

Example

Looking at what concepts relate to sociability and consideration traits in people

	Talk 1	Social Skills	Interest	Talk 2	Selfish	Liar
Talk 1	1.000					
Social Skills	.772	1.000				
Interest	.646	.879	1.000			
Talk 2	.074	-.120	.054	1.000		
Selfish	-.131	.031	-.101	.441	1.000	
Liar	.068	.012	.110	.361	.277	1.000

Factor 1

Factor 2

Selfishness is moderately correlated with Time spent talking about themselves.

Propensity to Lie is also moderately correlated with Time spent talking about themselves.

The correlation between propensity to lie and selfishness is a weak correlation but it is approaching moderate.

These are negative traits.

What is a Factor/Component?

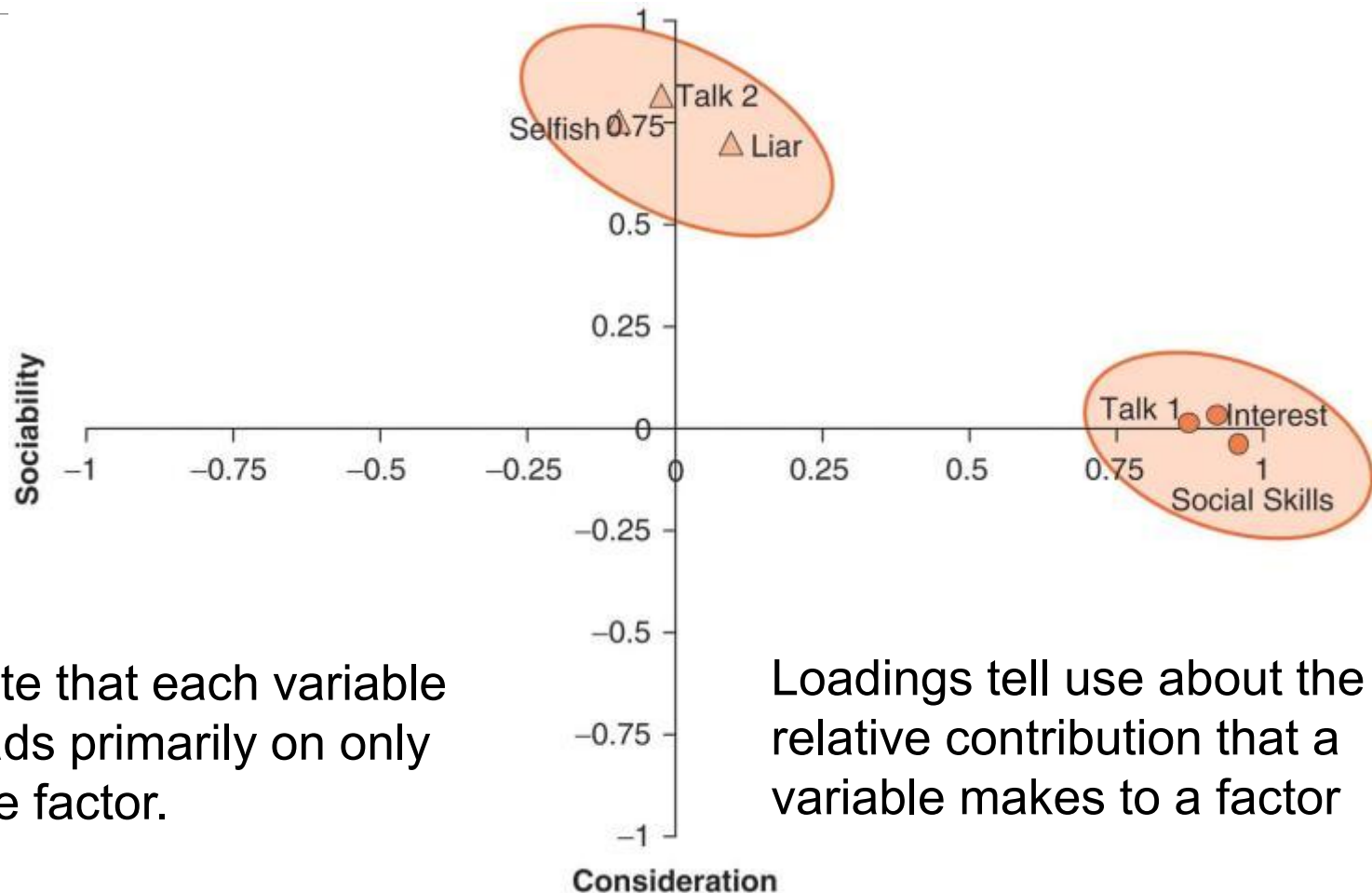
Factors/Component can be viewed as classification axes along which the individual variables can be plotted.

The greater the **loading** of variables onto a factor/component, the more the factor/component explains relationships among those variables.

- Loading can be viewed as the correlation between the latent variable and the manifest variable.

Ideally, variables should be strongly related to (or load onto) only one factor/component.

Graphical Representation of a factor plot



Note that each variable loads primarily on only one factor.

Loadings tell use about the relative contribution that a variable makes to a factor

Assumptions

A linear relationship exists between the latent variable and the manifest variables

$$F_i = b_1X_{1_i} + b_2X_{2_i} + \dots + b_nX_{ni} + \varepsilon_i$$

$$\text{Component}_i \text{ or } \text{Factor}_i = b_1\text{Variable}_{1i} + b_2\text{Variable}_{2i} + \dots + b_n\text{Variable}_{ni}$$

$$F_i = b_1X_{1_i} + b_2X_{2_i} + \dots + b_nX_{ni}$$

$$\begin{aligned} \text{Sociability}_i &= b_1\text{Talk1}_i + b_2\text{Social Skills}_i + b_3\text{Interest}_i \\ &\quad + b_4\text{Talk2}_i + b_5\text{Selfish}_i + b_6\text{Liar}_i \end{aligned}$$

$$\begin{aligned} \text{Consideration}_i &= b_1\text{Talk1}_i + b_2\text{Social Skills}_i + b_3\text{Interest}_i \\ &\quad + b_4\text{Talk2}_i + b_5\text{Selfish}_i + b_6\text{Liar}_i \end{aligned}$$

Mathematical Equation

$$F_i = b_1X_{1i} + b_2X_{2i} + \dots + b_nX_{ni} + \varepsilon_i$$

$$\text{Component}_i \dots \text{or} \dots \text{Factor}_i = b_1\text{Variable}_{1i} + b_2\text{Variable}_{2i} + \dots + b_n\text{Variable}_{ni}$$

Note that there is no b_0

The b values are called *factor loadings*

- **They can be interpreted as the correlation between the manifest variable and the factor**
- As with all correlations they range from -1 to +1

We should standardise our X s so that our interpretation of the levels of correlation make sense

Mathematical Equation

$$F_i = b_1X_{1_i} + b_2X_{2_i} + \dots + b_nX_{ni} + \varepsilon_i$$

$$\text{Component}_i \dots \text{or} \dots \text{Factor}_i = b_1\text{Variable}_{1_i} + b_2\text{Variable}_{2_i} + \dots + b_n\text{Variable}_{ni}$$

For every case in our dataset, we have the values of X but not of F

- Therefore, we cannot fit a regression line as we have done previously

But we can transform the relationships between the variables using a *method of extraction* to retrieve factors from the data

Loadings

Both FA and PCA are linear models in which **loadings** are used as weights.

- These **loadings** represent the **correlation** between a specific observed variable and a specific factor.
- Can be expressed as a matrix called the factor matrix or component matrix (if doing PCA).
- The assumption of FA (but not PCA) is that these algebraic factors represent real-world dimensions.

Variables	Factors	
	Sociability	Consideration
Talk 1	0.87	0.01
Social Skills	0.96	-0.03
Interest	0.92	0.04
Talk 2	0.00	0.82
Selfish	-0.10	0.75
Liar	0.09	0.70

Loadings

Loading=correlation between a specific observed variable and a specific factor.

Higher values mean a closer relationship. They are equivalent to standardised regression coefficients (β weights) in multiple regression.

Higher the value the better.

Variables	Factors	
	Sociability	Consideration
Talk 1	0.87	0.01
Social Skills	0.96	-0.03
Interest	0.92	0.04
Talk 2	0.00	0.82
Selfish	-0.10	0.75
Liar	0.09	0.70

There is no definition of what constitutes a high or low factor loading.
The convention is that a loading of magnitude above 0.3 (irrespective of sign) are considered high.

Example

In our example there are two factors underlying the *popularity* construct: **general sociability** and **(lack of) consideration**.

- We come up with the names

We can construct equations that describe each factor in terms of the variables that have been measured.

$$Sociability_i = b_1 \text{Talk } 1_i + b_2 \text{Social Skills}_i + b_3 \text{interest}_i + b_4 \text{Talk } 2 + b_5 \text{Selfish}_i + b_6 \text{Liar}_i + \varepsilon_i$$

$$(lack\ of)\ Consideration_i = b_1 \text{Talk } 1_i + b_2 \text{Social Skills}_i + b_3 \text{interest}_i + b_4 \text{Talk } 2 + b_5 \text{Selfish}_i + b_6 \text{Liar}_i + \varepsilon_i$$

Example

The values of the “ b ’s” in the two equations differ, depending on the relative importance of each variable to a particular factor.

$$\text{Sociability}_i = 0.87\text{Talk } 1_i + 0.96\text{Social Skills}_i + 0.92\text{Interest}_i + 0.00\text{Talk } 2_i - 0.10\text{Selfish}_i + 0.09\text{Liar}_i + \varepsilon_i$$

$$(\text{lack of}) \text{Consideration}_i = 0.01\text{Talk } 1_i - 0.03\text{Social Skills}_i + 0.04\text{interest}_i + 0.82\text{Talk } 2_i + 0.75\text{Selfish}_i + 0.70\text{Liar}_i + \varepsilon_i$$

Replace values of b with the co-ordinate of each variable on the graph.

Ideally, variables should have very high b -values for one factor and very low b -values for all other factors.

Principal Component Analysis (PCA)

A transformation of the data that estimates as many components as there are manifest variables

PCA reduces

- A set of *highly correlated* manifest variables
- To a *set of unrelated components*
- Arranged in descending order of importance

If PCA results in the first few components explaining a large amount of the variation we can without losing much information replace our manifest variables with our components.

Factor Analysis V Principal Components Analysis

FA attempts to achieve parsimony by explaining the maximum amount of *common variance* in a correlation matrix using the smallest number of explanatory constructs.

- These 'explanatory constructs' are called *factors*.

PCA tries to explain the maximum amount of *total variance* in a correlation matrix.

- It does this by transforming the original variables into a set of linear *components*.

What is the difference between a factor and a component?

Factor:

- A **factor** is a latent variable that is inferred to explain the underlying structure in the observed variables. It represents **shared variance** among variables, assumed to arise from common causes.

Purpose:

- **Factors** aim to uncover the hidden relationships or constructs that are not directly observable.

Interpretation:

- **Factors** are often interpreted as the theoretical constructs (e.g., intelligence, socioeconomic status) influencing the data.

Technique:

- Factor Analysis (Exploratory or Confirmatory)

Mathematical Basis:

- Factor analysis models the observed variables as linear combinations of a smaller number of factors plus unique error terms.
- $X = \Lambda F + \epsilon$
- where: X = observed data, Λ = loading matrix, F = factors, ϵ = unique variances (errors).

Common Uses:

- Psychology, social sciences, and any field where theoretical constructs are sought.

What is the difference between a factor and a component?

Component:

- A **component** is a linear combination of the original variables designed to maximize the variance explained in the dataset. Components are purely mathematical constructs.

Purpose:

- **PCA** aims to reduce dimensionality while preserving as much variance in the data as possible.

Interpretation:

- **Components** are linear transformations of the original variables without any assumption of underlying latent constructs.

Mathematical Basis:

- PCA identifies orthogonal (uncorrelated) axes that capture the maximum variance in the data sequentially. Technique: PCA
- $Z = WX$
- where: Z = principal components, W = weight matrix or eigenvectors of the covariance matrix, X = observed data.

Common Uses:

- Data preprocessing, exploratory data analysis, and visualization.

Conducting a dimension reduction: An Example (Andy Field)

Students often become stressed about statistics (RAQ) and the use of computers and/or R to analyze data.

A questionnaire has been developed to measure levels of stress about statistics, using computers and R

- see sample items on the following slides; the data can be found in RAQ.dat.

We want to investigate:

- Does the questionnaire measure a single construct?
- Or is it possible that there are multiple aspects comprising students' anxiety?

Companion File: MATH9102-W10-DimRed.qmd

SD = Strongly Disagree, D = Disagree, N = Neither, A = Agree, SA = Strongly

	SD	D
1 Statistics make me cry	☐	☐
2 My friends will think I'm stupid for not being able to cope with R	☐	☐
3 Standard deviations excite me	☐	☐
4 I dream that Pearson is attacking me with correlation coefficients	☐	☐
5 I don't understand statistics	☐	☐
6 I have little experience of computers	☐	☐
7 All computers hate me	☐	☐
8 I have never been good at mathematics	☐	☐
9 My friends are better at statistics than me	☐	☐
10 Computers are useful only for playing games	☐	☐
11 I did badly at mathematics at school	☐	☐
12 People try to tell you that R makes statistics easier to understand but it doesn't	☐	☐
13 I worry that I will cause irreparable damage because of my incompetence with computers	☐	☐
14 Computers have minds of their own and deliberately go wrong whenever I use them	☐	☐
15 Computers are out to get me	☐	☐
16 I weep openly at the mention of central tendency	☐	☐
17 I slip into a coma whenever I see an equation	☐	☐
18 R always crashes when I try to use it	☐	☐
19 Everybody looks at me when I use R	☐	☐
20 I can't sleep for thoughts of eigenvectors	☐	☐
21 I wake up under my duvet thinking that I am trapped under a normal distribution	☐	☐
22 My friends are better at R than I am	☐	☐

RAQ Questionnaire

TWENTY THREE QUESTIONS -
SOME OF THE QUESTIONS

Initial Considerations

The quality of the data ($GI \Rightarrow GO$).

Sample size is important!

- A sample of 300 or more will likely provide a stable factor solution but depends on the number of variables and factors identified.
- Sample to variable ratio $N:p$ where N is no of cases and p is no of variables aim for 10:1

The quality of analysis depends upon the quality of the data (GI $\Rightarrow$ GO).

Test variables should correlate quite well

- $r > .3$.

Avoid multicollinearity:

- several variables highly correlated, $r > .80$.

Avoid singularity:

- some variables perfectly correlated, $r = 1$ (other than with themselves)

Generate and Screen the correlation matrix, eliminate any variables that obviously cause concern.

Then Visualize the correlation matrix.

Step 1 : Screen Correlation Matrix

Check that there is enough variance of the right kind

Generate statistics to:

1. Check that your variables are sufficiently correlated for dimension reduction (Barlett's Test of Sphericity)
2. Check that there is enough variance in your variables that could be caused by an underlying factor/dimension (Kaiser-Meyer Olsen)
3. Check for collinearity and redundancy (Determinant)

Step 2: Check if the data is suitable for dimension reduction

Step 2a:
Check that
your variables
are
sufficiently
correlated for
dimension
reduction.

Use The Bartlett Test of Sphericity.

- Compares the correlation matrix with a matrix of zero correlations
 - technically called the identity matrix, which consists of all zeros except the 1's along the diagonal
- If Bartlett's test is statistically significant it suggests that the variables are correlated and a PCA/FA would be useful
 - i.e. You are highly unlikely to have obtained the observed correlation matrix from a population with zero correlation.
 - Bartlett, M. S. (1950). Tests of significance in factor analysis. *British journal of psychology*.

Step 2b:
Check that
there is
enough
variance in
your variables
that could be
caused by an
underlying
factor/dimen
sion.

Use the Kaiser-Meyer-Olkin Measure of Sampling Adequacy (MSA) test.

- Doesn't produce a p value
- High values close to 1 suggest that PCA/FA might be useful
- Values over 0.8 or over are considered strong
- Anything less than 0.5 suggests that PCA/FA won't be useful

Kaiser put together the following descriptors:

- 0.00 to 0.49 unacceptable.
- 0.50 to 0.59 miserable.
- 0.60 to 0.69 mediocre.
- 0.70 to 0.79 middling.
- 0.80 to 0.89 meritorious.
- 0.90 to 1.00 marvellous.
- Source: Kaiser, Henry F., and John Rice. "Little jiffy, mark IV." Educational and psychological measurement 34.1 (1974): 111-117.

Step 2c: Check for collinearity and redundancy

Check if it will be difficult to extract factors as they will not be distinct enough.

Calculate the determinant of the matrix (correlation or covariance)

- A measure of the spread or volume of the data in the original space.
- A zero or near-zero determinant indicates that the data lie on a lower-dimensional subspace (with little or no variance in certain directions).
- A larger determinant suggests that the data have significant variance across many dimensions, which can be reduced in the dimension reduction process (e.g., PCA) to retain the most important features.
- Inspect determinant and check to see if it is greater than 0.00001.
- Important for factor analysis not for PCA
- Field, Miles and Field (2016)

Considerations – Is dimension reduction possible?

Check the Determinant:

- Indicator of multicollinearity
- Should be greater than 0.00001

Check Kaiser–Meyer–Olkin (KMO):

- Measures sampling adequacy (how well suited is your data to the dimension reduction)
- Should be greater than .5
- Indicates the proportion of variance in your variables that might be caused by an underlying factor.

Check Bartlett's test of Sphericity:

- Tests whether the R-matrix is an identity matrix
- Should be significant at $p < .05$
- We can see the correlations ourselves, this just confirms that it is possible to combine the variables because they are correlated.

Step 2d: Report

Bartlett's test of sphericity revealed a significant result, $\chi^2(253) = 19107.61$, $p < .001$, indicating that the correlation matrix is not an identity matrix and thus suitable for factor analysis.

Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy was 0.93, indicating that the data is highly suitable for factor analysis (values above 0.90 are considered excellent).

Determinant of the correlation matrix was 0.00057, suggesting that the data is suitable for factor analysis

Reporting the Results

See example [MATH9102-W10-DimRed.html](#)

Time to get started

Download the code and data for this week – MATH9102-W10-DimRed.qmd and the .html output.

You are going to work with the dataset **wine.csv**

Download the Quarto file **MATH9102-W10-DimRedExercise.qmd**

- This sets up the data and includes instructions for all steps – you just need to write the code.
- At this point just do **TASK ONE**

Refer to MATH9102-W10-DimRed.qmd for sample code.

Step 3: Do The Dimension Reduction

RAQ

- In this case we are using all the variables, and we expect components
- We are going to use Principal Component Analysis first
- We also create a Scree plot which we will use later

FA/PCA

Principal Components Analysis (PCA)

- This is a data reduction method, not a true factor analysis (as it analyses all of the variance of the variables) so attempts to extract same number of components as there are variables.

Factor Analysis

- Need to choose a method of factor extraction
- There are six extraction methods : unweighted least squares, generalized least squares, maximum likelihood, principal axis factoring, alpha factoring and image factoring
- **The general rule of thumb is:**
 - If your data are normally distributed then use **Maximum Likelihood (ML)**
 - If your data are not normally distributed then use **Principal Axis Factoring (PAF)**

Step 3: Do the Dimension Reduction

```
#Principal Component Analysis
```

```
#On raw data using principal components analysis,  
nfactors is set to the number of variables we expect to  
get out which, in this case, is equal to the number  
going in.
```

```
pc1 <- principal(raqData, nfactors = length(raqData),  
rotate = "none")
```

```
pc1 # Output all the details
```

```
plot(pc1$values, type = "b") # Generate a scree plot
```

Step 4: Decide how many factors/components to retain

How many latent variables do we want to keep (retain)?

- The common method is to select factors with *Eigenvalues greater than 1* option

Eigenvalue = the *variance* of a component or factor

- Eigenvalues shows variance explained by that particular factor out of the total variance

Kaiser (1960) recommends retaining all factors with eigenvalues greater than 1.

- Based on the idea that eigenvalues represent the amount of variance explained by a factor and that an eigenvalue of 1 represents a substantial amount of variation.
- Jolliffe (1972; 1986) reported that Kaiser's criterion is too strict and recommended retaining factors with eigenvalues more than 0.7.

Extracting the components

If we have n manifest variables in our analysis, PCA transforms the data such that the total variance of the components n will be redistributed among the components

- The first will have the largest eigenvalue
- The next the next highest etc.

If the eigenvalue is divided by n the result is the proportion of variance explained by that factor

A rule of thumb is to retain in the final solution all factors with an Eigenvalue greater than one

Retrieving our eigenvalues

```
pc1$values #output eigenvalues
```

```
#Another way to look at Eigen values plus  
variance explained
```

```
pcf=princomp(raqData)
```

```
factoextra::get_eigenvalue(pcf)
```

```
pc1$values #output eigenvalues
```

```
## [1] 7.2900471 1.7388287 1.3167515 1.2271982 0.9878779 0.8953304 0.8055604  
## [8] 0.7828199 0.7509712 0.7169577 0.6835877 0.6695016 0.6119976 0.5777377  
## [15] 0.5491875 0.5231504 0.5083962 0.4559399 0.4238036 0.4077909 0.3794799  
## [22] 0.3640223 0.3330618
```

```
#Another way to look at eigen values plus variance explained  
pcf=princomp(raqData)  
factoextra::get_eigenvalue(pcf)
```

	eigenvalue	variance.percent	cumulative.variance.percent
## Dim.1	7.0430039	31.059169	31.05917
## Dim.2	1.9919543	8.784383	39.84355
## Dim.3	1.3755696	6.066168	45.90972
## Dim.4	1.1415644	5.034221	50.94394
## Dim.5	1.0453716	4.610018	55.55396
## Dim.6	0.8929067	3.937658	59.49162
## Dim.7	0.8530615	3.761943	63.25356
## Dim.8	0.8136079	3.587955	66.84152
## Dim.9	0.7489525	3.302830	70.14435
## Dim.10	0.7054281	3.110890	73.25524
## Dim.11	0.6894466	3.040413	76.29565
## Dim.12	0.6195139	2.732014	79.02766
## Dim.13	0.5624056	2.480171	81.50783
## Dim.14	0.5500250	2.425573	83.93341
## Dim.15	0.5300000	2.341170	86.27458

Our example

Finding Factors

Cattell (1966) suggests using the 'point of inflexion' of the scree plot to decide how many factors to extract

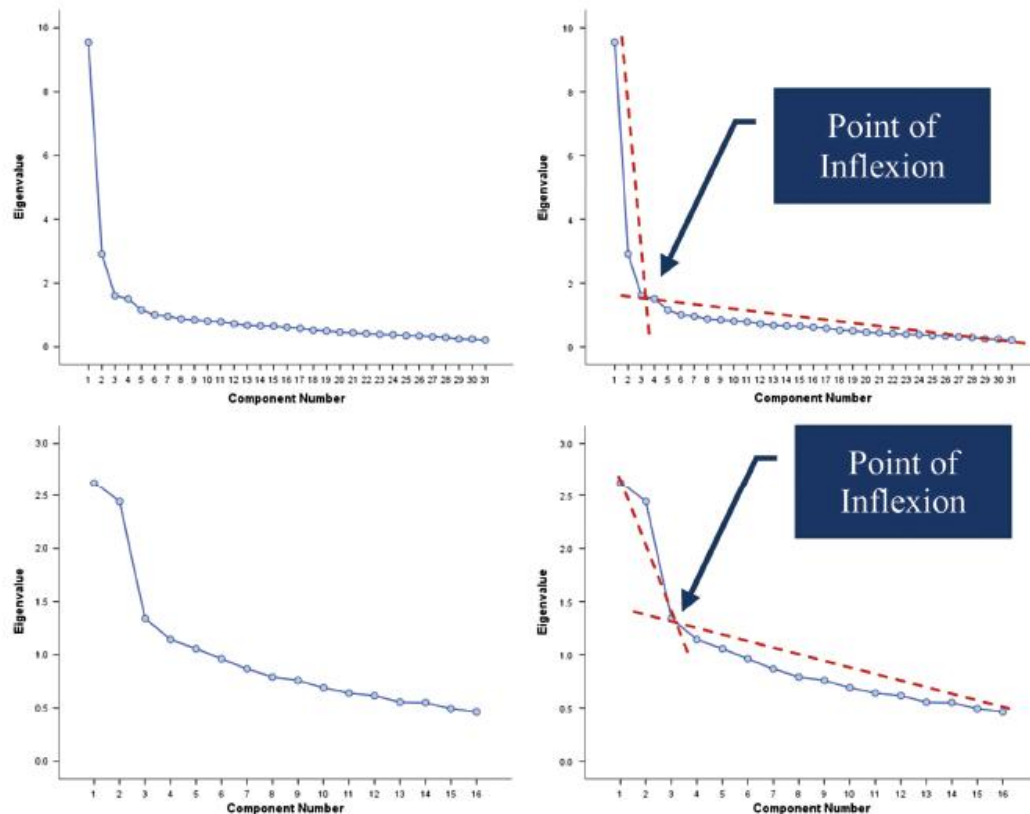
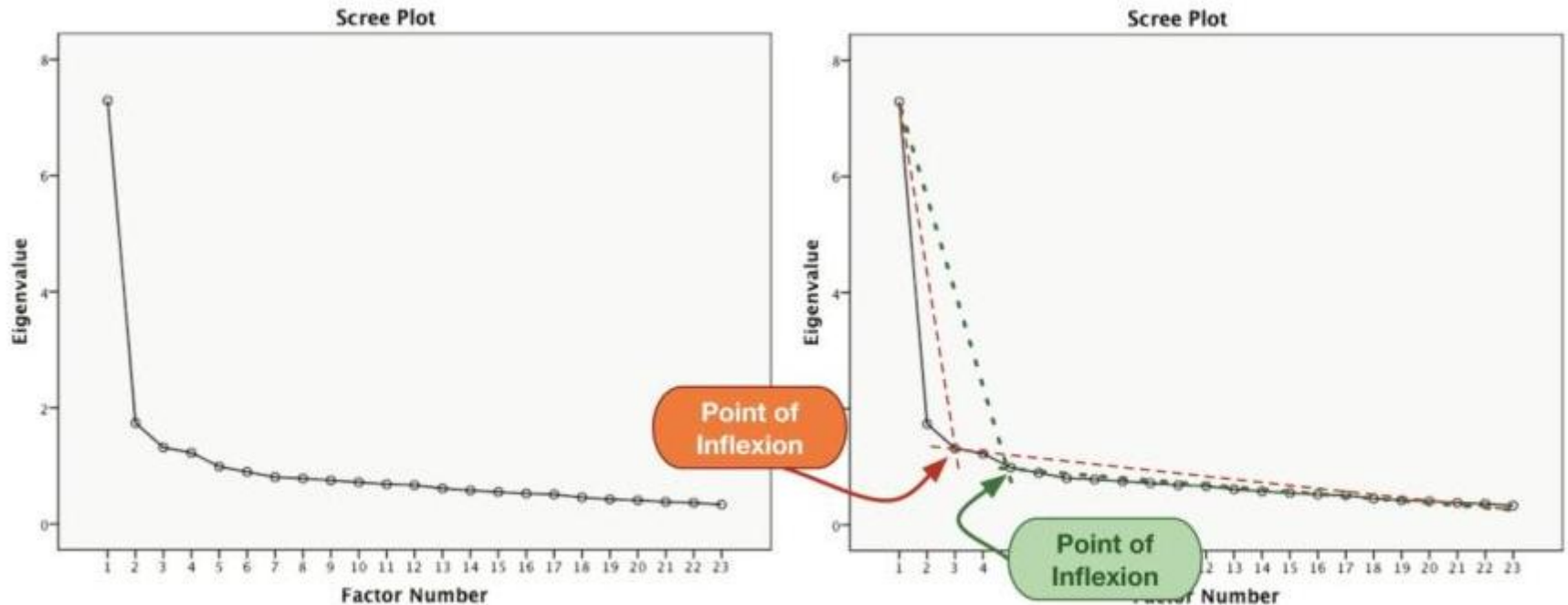


FIGURE 17.3 Examples of scree plots for data that probably have two underlying factors

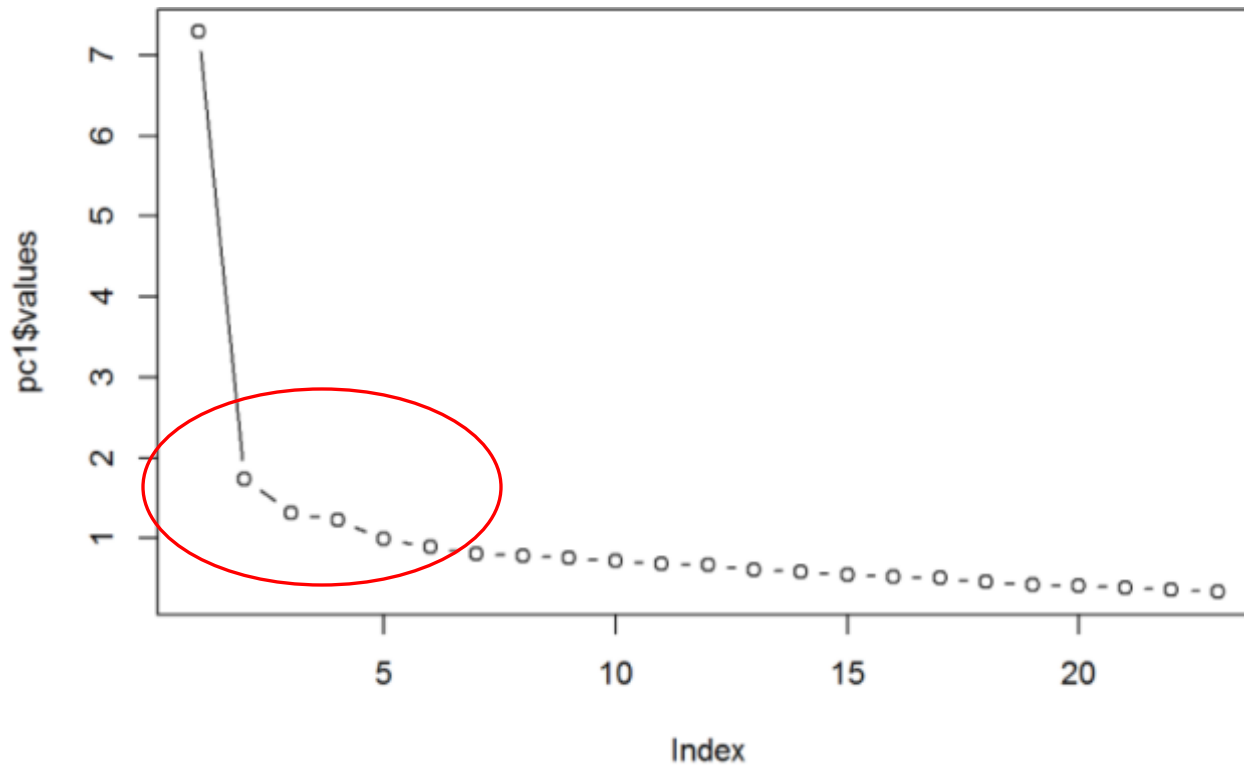
Scree Plot



OUTPUT 17.6

Look for change in shape of the plot – points of inflexion, only factors/components above this point are retained

Our Example



Factor/Component Extraction

Which Rule?

- Use Kaiser's Extraction when
 - less than 30 variables, communalities after extraction > 0.7 .
 - sample size > 250 and mean communality ≥ 0.6 .
- Scree plot is good if sample size is > 200 .

Total Variance Explained

Indicates how much of the variability in the data has been modelled by the extracted factors

Variance Explained

##	eigenvalue	variance.percent	cumulative.variance.percent
## Dim.1	7.0430039	31.059169	31.05917
## Dim.2	1.9919543	8.784383	39.84355
## Dim.3	1.3755696	6.066168	45.90972
## Dim.4	1.1415644	5.034221	50.94394
## Dim.5	1.0453716	4.610018	55.55396
## Dim.6	0.8929067	3.937658	59.49162
## Dim.7	0.8530615	3.761943	63.
## Dim.8	0.8136079	3.587955	66.
## Dim.9	0.7489525	3.302830	70.
## Dim.10	0.7054281	3.110890	73.
		3.040413	76.
		2.732014	79.
		2.480171	81.
		2.425573	83.
		2.341270	86.

Calculated by
dividing Eigenvalue
by the number of
factors/components

Indicates how much of the
variability in the data has been
modelled by the extracted
factors.

Factor/Component 11 explains
31.69%,
Factor/Component 1+2
explain 39.26% etc.

Together the 4 uncorrelated
factors summarise 50.32% of
the information in the original
inter-related variables

In R

```
pc1$Vaccounted#Variance  accounted  
for
```

Step 4b: Examine the Communalities

b is the *correlation* between the manifest variable and the factor or component

The square of the correlation is the proportion of *variance in the manifest variable that can be explained by the factor*

If we square and sum all the factor loadings for a manifest variable across the factor equations we have the communality

- **How much of the variance in the manifest variable is accounted for by the combination of factors**

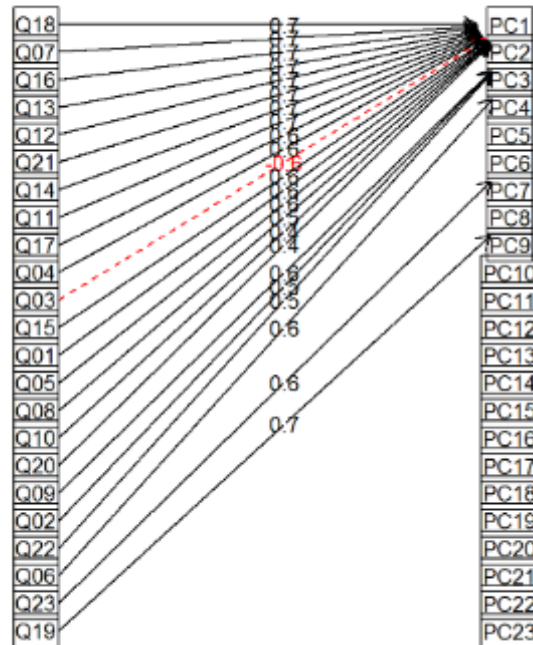
Deciding on Factors/Components to Retain

Need to look at both loadings and communalities

Strong data

- Uniformly high communalities (>0.8 is strong but it is more likely to be in the region of 0.4-0.7).
 - Item loadings above 0.3 is acceptable.
- None, or few, cross loadings (a variable loads higher than 0.32 on two or more factors)
- Several variables loading on to each factor/component (factor loadings of 0.3 or greater).
- A factor/component with fewer than 3 items is typically weak.

Components Analysis



Loadings (PA1=Dim1=Component 1)

Communalities

In the initial PCA because we have as many components/factors as manifest variables this will be 1

If we drop components it will be less than 1

- This will then provide an indication of how much each manifest variable contributes to our reduced solution.

Factor/Component Matrix - Loadings

	Factor			
	1	2	3	4
Statistics makes me cry	.557	.112	-.090	.205
My friends will think I'm stupid for not being able to cope with SPSS	-.281	.372	.185	.097
Standard deviations excite me	-.605	.249	.204	-.050
I dream that Pearson is attacking me with correlation coefficients	.607	.082	-.046	.204
I don't understand statistics	.522	.039	-.011	.157
I have little experience of computers	.552	.023	.489	-.223
All computers hate me	.663	-.025	.221	.013
I have never been good at mathematics	.546	.483	-.282	-.188
My friends are better at statistics than me	-.266	.460	.142	.190
Computers are useful only for playing games	.404	-.007	.157	-.094
I did badly at mathematics at school	.646	.313	-.219	-.257
People try to tell you that SPSS makes statistics easier to understand but it doesn't	.643	-.099	.070	.160
I worry that I will cause irreparable damage because of my incompetence with computers	.650	.022	.212	-.078
Computers have minds	.628	-.037	.163	.051

Correlation between a specific manifest/observed variable and a specific factor
These are the b's for the factor equations

Called component matrix for PCA

Communalities

How much variance in the manifest/observed variables is accounted for by the factors found

	Communalities	
	Initial	Extraction
Statistics makes me cry	.373	.373
My friends will think I'm stupid for not being able to cope with SPSS	.188	.260
Standard deviations excite me	.398	.472
I dream that Pearson is attacking me with correlation coefficients	.385	.419
I don't understand statistics	.291	.299
I have little experience of computers	.427	.594
All computers hate me	.470	.489
I have never been good at mathematics	.490	.646
My friends are better at statistics than me	.220	.339
Computers are useful only for playing games	.197	.197
I did badly at mathematics at school	.530	.629
People try to tell you that SPSS makes statistics easier to understand but it doesn't	.424	.453

Communality = sum of the square of the loading of a variable across a component/factor

Communalities

Factor Matrix^a

	1	2	3	4
I wake up under my duvet thinking that I am trapped under a normal distribution	0.594	0.264	0.149	0.15

Communalities

	Extraction
I wake up under my duvet thinking that I am trapped under a normal distribution	.467

I wake up under my duvet thinking I
am trapped under a normal
distribution = $.594^2 + .264^2 + .149^2 + .15^2 = .467$

Step 5: Rotation

Given the communalities, there are multiple solutions for loadings that result in the same communality and correlation

- “Correct” answer is dependent upon interpretability

To aid interpretation it is possible to maximise the loading of a variable on one factor while minimising its loading on all other factors

The aim of rotation is to clarify the data structure

- You cannot apply a rotation if only one factor emerges
- Aim to have non-zero loadings on each factor for only some of the variables

Rotation reallocates the variable variance among the factors

- Shifts variance from earlier factors to later ones
- Interpretation shifts from unrotated factors delineating the most comprehensive data patterns to factors delineating the distinct groups of interrelated data.

Rotation – Why?

The factor patterns define decreasing amounts of variation in the data.

Each pattern may involve all or most of the variables and the variables may have moderate or high loadings for several factor patterns

To uncover the first pattern, a factor is fitted to the data to account for the greatest regularity;

- Each successive factor is fitted to best define the remaining regularity.
- *The result of this is that the first unrotated factor may be located between independent clusters of interrelated variables.*
- These clusters cannot be distinguished in terms of their loadings on the first factor, although they will have loadings different in sign on the second and subsequent factors
- Each factor is rotated until it defines a distinct cluster of interrelated variables.

Geometric Rotation

Factor extraction equivalent to coordinate planes

Factors are the axes

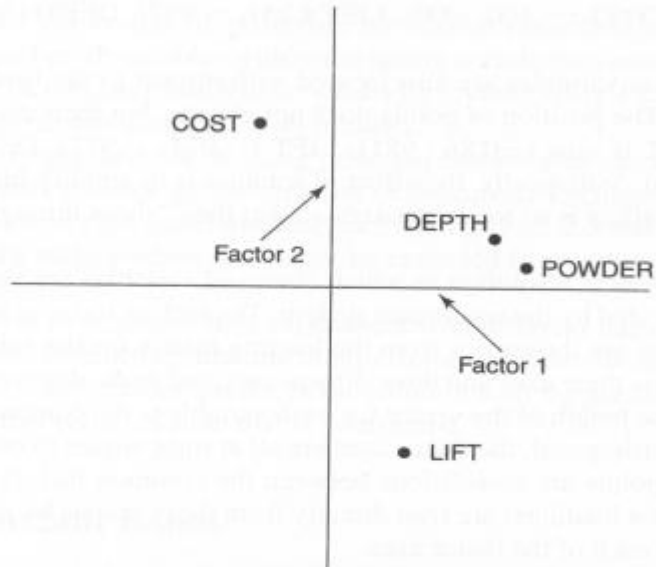
Length of the line from the origin to the variable coordinates is equal to the communality for that variable

Orthogonal Factors are at right angles

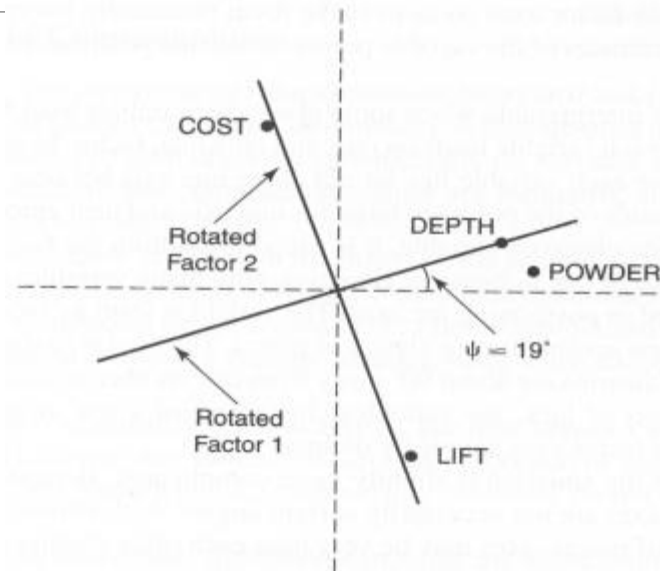
Factor loadings are found by dropping a line from the variable coordinates to the factor at a right angle

Repositioning the axes changes the loadings on the factor but keeps the relative positioning of the points the same

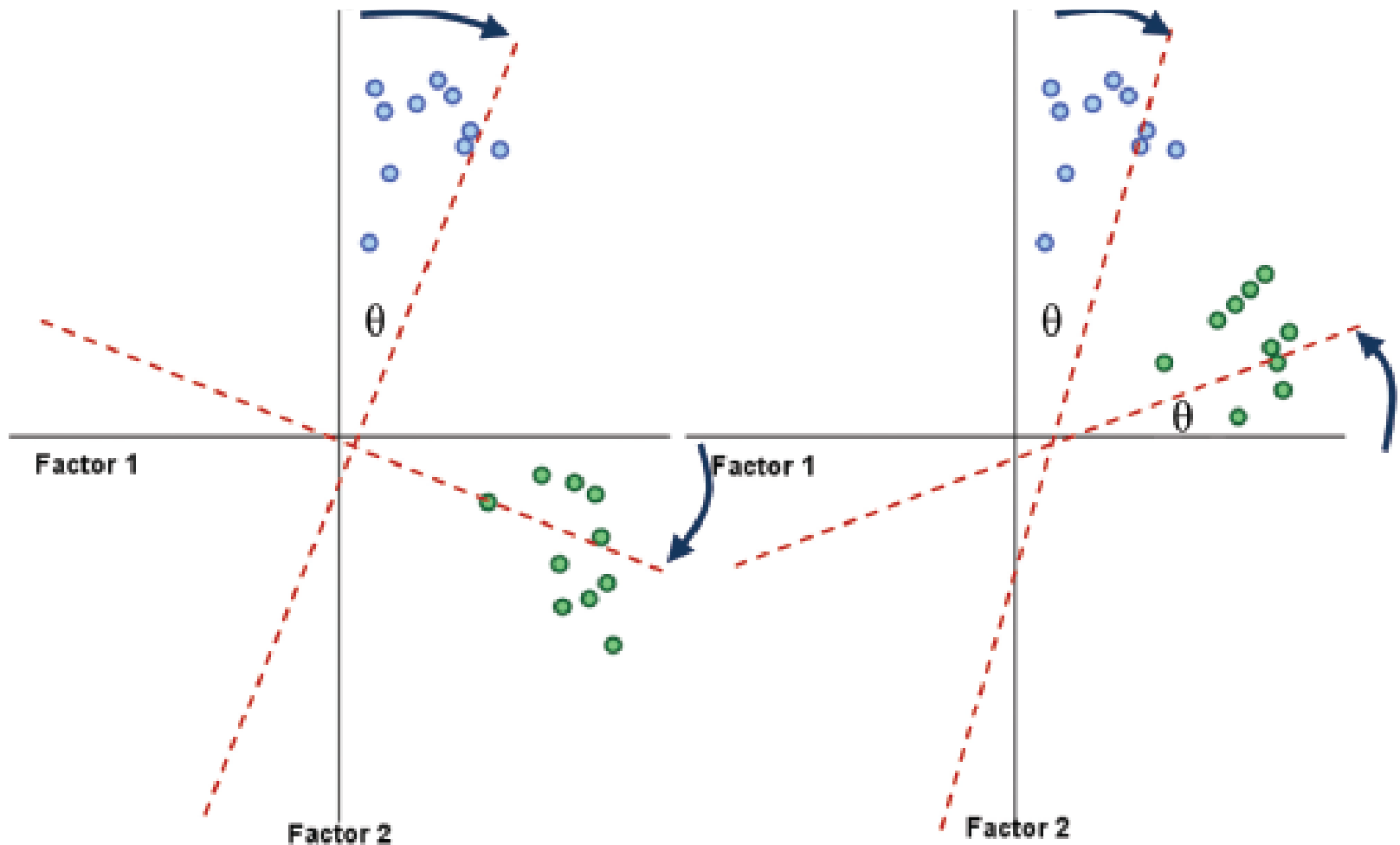
Geometric Rotation



(a) Location of COST, LIFT, DEPTH, and POWDER after extraction, before rotation



(b) Location of COST, LIFT, DEPTH, and POWDER vis-à-vis rotated axes



Orthogonal Rotation

Oblique Rotation

Rotating the axes - Aim is to make the location of the axes fit the actual data points better. Move to a position that encompasses the actual data points better overall. This will hopefully make them more interpretable.

Rotation

```
#Rotation PCA
```

```
pc2 <- principal(raqData, nfactors  
= 4, rotate = "varimax") #Extracting  
4 factors
```

```
print.psych(pc2, cut = 0.3, sort =  
TRUE)
```

```
#Get the output as you would for  
initial solution
```

Rotated Factor Matrix(FA)

Check the factor loadings after rotation

Need to consider rotated
factor loadings before making
a final decision

Reporting the Results PCA

See example [MATH9102-W10-DimRed.html](#)

Step 3: Factor analysis

```
#Factor Analysis

#Principal Axis Factoring fm=ml for normally
distributed data, pa for non normally distributed

factsol <- fa(raqMatrix, nfactors=4, obs=NA, n.iter=1,
rotate="varimax", fm="ml")

psych::print.psych(factsol, cut=0.3, sort=TRUE)

fa.graph(factsol)

fa.sort(factsol$loading)

fa.diagram(factsol) #create a diagram showing the
factors and how the manifest variables load

plot(factsol$values, type = "b") #scree plot
```

Time for you to try

Refer to MATH9102-W10-DimRed.qmd for sample code and the .html for output

Using **MATH9102-W10-DimRedExercise.qmd**:

Do Task TWO – Conduct a PCA

Time for you to try

Refer to MATH9102-W10-DimRed.qmd for sample code and the .html for output

Using **MATH9102-W10-DimRedExercise.qmd**:

Do Task THREE – Conduct an EFA

Step 6: Reliability Analysis (For Both)

A Cronbach's alpha measurement of consistency tells you how consistently the variables behave as a scale.

- Internal consistency refers to the general agreement between multiple items (often Likert scale items) that make-up a composite score of a survey measurement of a given construct.
- This agreement is generally measured by the correlation between items.

If it's high (say .80 or .90), then we probably have one factor/component

Interpreting Cronbach's Alpha

Depends on the number of items

- More questions = bigger α
- Low numbers of items (<10) can result in low levels of alpha

Treat Subscales separately

Remember to reverse score reverse phrased items!

- If not, α is reduced and can even be negative

Reliability:

Cronbach's Alpha

- Ranges from 0 (no reliability) to 1 (complete reliability)
- Should be .7 or greater to be considered “reliable”.
- The closer Cronbach's alpha coefficient is to 1.0 the greater the internal consistency of the items in the scale.
- George and Mallery provide the following rules of thumb:
 - > .9 – Excellent
 - > .8 – Good
 - > .7 – Acceptable
 - > .6 – Questionable
 - > .5 – Poor and
 - < .5 – Unacceptable

George and Mallery, SPSS for Windows step by step: A simple guide and reference. 11.0 update (4th ed.) (2011)

Item-Total Statistics: Fear of Computers sub-scale

```
raw_alpha std.alpha G6(smc) average_r S/N    ase mean    sd median_r
      0.82      0.82      0.81      0.4 4.6 0.0052   3.4 0.71      0.39
```

```
lower alpha upper      95% confidence boundaries
0.81 0.82 0.83
```

Reliability if an item is dropped:

```
raw_alpha std.alpha G6(smc) average_r S/N alpha se  var.r med.r
Q06      0.79      0.79      0.77      0.38 3.7   0.0063 0.0081  0.38
Q07      0.79      0.79      0.77      0.38 3.7   0.0063 0.0079  0.36
Q10      0.82      0.82      0.80      0.44 4.7   0.0053 0.0043  0.44
Q13      0.79      0.79      0.77      0.39 3.8   0.0062 0.0081  0.38
Q14      0.80      0.80      0.77      0.39 3.9   0.0060 0.0085  0.36
Q15      0.81      0.81      0.79      0.41 4.2   0.0056 0.0095  0.44
Q18      0.79      0.78      0.76      0.38 3.6   0.0064 0.0058  0.38
```

Item statistics

```
      n raw.r std.r r.cor r.drop mean    sd
Q06 2571  0.75  0.74  0.68  0.62  3.8 1.12
Q07 2571  0.75  0.73  0.68  0.62  3.1 1.10
Q10 2571  0.54  0.57  0.44  0.40  3.7 0.88
Q13 2571  0.72  0.73  0.67  0.61  3.6 0.95
Q14 2571  0.70  0.70  0.64  0.58  3.1 1.00
Q15 2571  0.64  0.64  0.54  0.49  3.2 1.01
Q18 2571  0.76  0.76  0.72  0.65  3.4 1.05
```

Alpha above .7 is
strong, .6 is
acceptable

Item-Total Statistics: Fear of Statistics sub-scale

raw_alpha	std.alpha	G6(smc)	average_r	S/N	ase	mean	sd	median_r
0.61	0.64	0.71	0.18	1.8	0.01	3.1	0.5	0.34

lower	alpha	upper	95% confidence boundaries
0.59	0.61	0.63	

Reliability if an item is dropped:

	raw_alpha	std.alpha	G6(smc)	average_r	S/N	alpha	se	var.r	med.r
Q01	0.52	0.56	0.64	0.15	1.3	0.0128	0.123	0.33	
Q03	0.80	0.80	0.79	0.37	4.1	0.0061	0.007	0.40	
Q04	0.50	0.55	0.64	0.15	1.2	0.0133	0.119	0.33	
Q05	0.52	0.57	0.66	0.16	1.3	0.0127	0.129	0.33	
Q12	0.52	0.56	0.65	0.15	1.3	0.0131	0.120	0.33	
Q16	0.51	0.55	0.63	0.15	1.2	0.0133	0.116	0.33	
Q20	0.56	0.60	0.68	0.18	1.5	0.0118	0.133	0.39	
Q21	0.50	0.55	0.63	0.15	1.2	0.0136	0.117	0.30	

Look for items that
would improve
alpha if removed

In R

```
#Reliability analysis
```

```
computerFear<-raqData[,c(6, 7, 10, 13, 14, 15, 18)]
```

```
statisticsFear <- raqData[, c(1, 3, 4, 5, 12, 16, 20, 21)]
```

```
mathFear <- raqData[, c(8, 11, 17)]
```

```
peerEvaluation <- raqData[, c(2, 9, 19, 22, 23)]
```

```
alpha(computerFear)
```

#If some items are to be reversed keyed, then either specify the direction of all items or just a vector of which items to reverse

```
alpha(statisticsFear, keys = c(1, -1, 1, 1, 1, 1, 1, 1))
```

```
alpha(mathFear)
```

```
alpha(peerEvaluation)
```

```
alpha(statisticsFear) #for illustrative purposes
```

Time for you to try

Refer to MATH9102-W10-DimRed.qmd for sample code and the companion .html for the output.

Using MATH9102-W10-DimRedExercise.qmd

Do Task FOUR – Reliability Analysis for both PCA and EFA (code is included –just interpret it)

The End?

Describe Factor Structure/Reliability

What items should be retained?

- What items did you eliminate and why?

Application

- How does it fit in with your theory?
- Where will your questionnaire be used? (if you are this is your research focus)

Exploratory Factor Analysis (EFA)

Is a variable reduction technique which identifies the number of latent constructs and the underlying factor structure of a set of variables

Hypothesizes an underlying construct, a variable not measured directly

Estimates factors which influence responses on observed variables

Allows you to describe and identify the number of latent constructs (factors)

Includes unique factors, error due to unreliability in measurement

Exploratory Factor Analysis (EFA)

Traditionally has been used to explore the possible underlying factor structure of a set of measured variables without imposing any preconceived structure on the outcome

Factor loadings (parameter estimates) help interpret factors.

Loadings are the correlation between observed variables and factors

- are standardized regression weights if variables are standardized (weights used to predict variables from factor) and are path coefficients in path analysis.

Standardized linear weights represent the effect size of the factor on variability of observed variables.

Principal Component Analysis (PCA)

Is a large sample procedure

Is a variable reduction technique

Is used when variables are highly correlated

Reduces the number of observed variables to a smaller number of principal components which account for most of the variance of the observed variables

The total amount of variance in PCA is equal to the number of observed variables being analyzed.

In PCA, observed variables are standardized,

- e.g., mean=0, standard deviation=1, diagonals of the matrix are equal to 1.

PCA

Not based on any causative relationship between the components and the manifest variables.

It is not proposing that the components are latent variables

It is simply rearranging the data under the given constraints.

PCA and EFA Requirements

Measurement scale is interval or ratio level (or ordinal)

Random sample

- At least 5 observations per observed variable and at least 100 observations in total required.

Larger sample sizes are recommended for more stable estimates, 10-20 observations per observed variable

Over sample to compensate for missing values

Linear relationship between observed variables

Factor Analysis v Principal Component Analysis

Both attempt to produce a smaller number of linear combinations of original variables that captures or accounts for most of the variability in the pattern of correlations

In PCA

- Assumes all the variance is common
- There is little unique or error variance – it is a very small part of the total
- Objective is to identify minimum number of factors to capture the total amount of variation

In FA

- Factors are estimated using a mathematical model
- Only the shared/common variance is used
- Objective is to identify latent constructs represented in the original variables

Factor Analysis v Principal Component Analysis

Results from PCA and EFA may be similar

- When number of variables exceeds 30 or
- Communalities exceed .6 for most variables

Confirmatory factor analysis

- Not an exploratory technique
- Requires hypotheses regarding factors and variables

Procedure for factor analysis & PCA

